

Laws of Motion

Comprehensive Physics Notes & Formula Sheet for CBSE Class 11 & JEE Mains/Advanced

CBSE BOARDS

JEE MAINS

JEE ADVANCED

QUICK REVISION

1. Fundamental Concepts: Force & Inertia

Force: An external agency (push or pull) that changes or tends to change the state of rest, uniform motion, or direction of a body. It is a vector quantity with SI unit **Newton (N)** [$1\text{ N} = 1\text{ kg}\cdot\text{m/s}^2$]. Dimensional Formula: $[M L T^{-2}]$.

Inertia: The inherent property of a material body by virtue of which it resists any change in its state of rest or uniform motion in a straight line. Inertia is directly proportional to mass ($\text{Inertia} \propto m$).

- **Inertia of Rest:** Resistance to start moving (e.g., passengers jerk backward when a bus accelerates).
- **Inertia of Motion:** Resistance to stopping (e.g., passengers jerk forward when brakes are applied).
- **Inertia of Direction:** Resistance to changing direction (e.g., mud thrown off rotating tires).

2. Newton's Laws of Motion

First Law (Law of Inertia)

Every body continues in its state of rest or uniform motion in a straight line unless acted upon by a net external force.

Second Law (Law of Force & Acceleration)

The rate of change of linear momentum of a body is directly proportional to the applied net external force and takes place in the direction of the force.

$$\text{Linear Momentum: } \mathbf{p} = m\mathbf{v} \quad | \quad \text{Force: } \mathbf{F}_{\text{net}} = \frac{d\mathbf{p}}{dt} = \frac{d(m\mathbf{v})}{dt}$$

If mass m is constant:

$$\mathbf{F}_{\text{net}} = m(\frac{d\mathbf{v}}{dt}) = m\mathbf{a} \Rightarrow F_x = m a_x, F_y = m a_y, F_z = m a_z$$

Third Law (Action and Reaction)

To every action, there is always an equal and opposite reaction. Forces always exist in pairs.

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA} \quad (\text{Force on A by B} = - \text{Force on B by A})$$

KEY RULES FOR ACTION-REACTION PAIRS (BOARD/JEE CONCEPTUAL)

- Action and reaction act simultaneously on **two different objects**, never on the same object. Therefore, they never cancel each other out.
- They are of the same nature (e.g., gravitational-gravitational, contact-contact).

3. Impulse & Linear Momentum

Impulse (J): The total effect of a large force acting over a very short time interval.

$$J = \int F dt = F_{avg} \Delta t = \Delta p = p_{final} - p_{initial}$$

Impulse-Momentum Theorem: The impulse of a force is equal to the change in linear momentum of the body. Area under the $F - t$ graph gives Impulse.

JEE TIP: LAW OF CONSERVATION OF LINEAR MOMENTUM

If net external force on a system is zero ($\Sigma F_{ext} = 0$), then total momentum remains conserved ($P_{initial} = P_{final}$).

Recoil Velocity of Gun: $v_{recoil} = - (m / M) u$

4. Free Body Diagrams (FBD) & Common Forces

To solve mechanics problems, isolate the body and draw all external forces acting on it:

- **Weight ($W = mg$):** Acts vertically downward towards earth center.
- **Normal Force (N):** Electromagnetic contact force acting perpendicular to the surface of contact.
- **Tension (T):** Pulling force transmitted through a string/rope, directed away from the body along the string.
- **Spring Force (F_s):** Restoring force $F_s = -kx$ (Hooke's Law).

Standard Systems & Formulas

System Type	Setup / Diagram Description	Formulae (Acceleration & Tension)
Atwood Machine	Two masses ($m_1 > m_2$) over light frictionless pulley	$a = [(m_1 - m_2) / (m_1 + m_2)] g$ $T = [2 m_1 m_2 / (m_1 + m_2)] g$
Masses in Contact	Two blocks m_1, m_2 on horizontal smooth floor pushed by F	$a = F / (m_1 + m_2)$ Contact Force: $N = [m_2 / (m_1 + m_2)] F$
Block on Smooth Incline	Mass m on smooth plane inclined at angle θ	$a = g \sin\theta$ Normal reaction: $N = mg \cos\theta$

5. Apparent Weight in a Lift (Elevator)

Motion Condition	Apparent Weight (N)	Sensation
Lift at rest or moving with uniform velocity ($a = 0$)	$N = mg$	Normal weight
Accelerating upwards with a	$N = m(g + a)$	Heavier
Accelerating downwards with a	$N = m(g - a)$	Lighter
Free fall ($a = g$, cable snaps)	$N = 0$	Weightlessness

6. Friction

Contact force that opposes relative motion (or tendency of relative motion) between two surfaces.

- **Static Friction (f_s):** Self-adjusting force. $0 \leq f_s \leq f_{s,max}$.
- **Limiting Friction ($f_{s,max}$):** Maximum static friction force. $f_{s,max} = \mu_s N$.

- **Kinetic Friction (f_k):** Opposes actual relative motion. $f_k = \mu_k N$ (where $\mu_k < \mu_s$).

$$\text{Angle of Friction } (\lambda): \tan \lambda = \mu \quad | \quad \text{Angle of Repose } (\theta): \tan \theta = \mu_s$$

7. Circular Motion & Vehicle Dynamics

Centripetal acceleration required for motion in circle of radius r : $a_c = v^2 / r = \omega^2 r$.

Key Maximum Safety Speeds

- **Car on Unbanked Level Circular Road:** Centripetal force supplied solely by static friction.

$$v_{max} = \sqrt{(\mu_s g r)}$$

- **Car on Banked Circular Road (Without Friction):**

$$v_{opt} = \sqrt{(r g \tan \theta)} \Rightarrow \tan \theta = v^2 / (r g)$$

- **Car on Banked Circular Road (With Friction):**

$$v_{max} = \sqrt{[r g (\mu_s + \tan \theta) / (1 - \mu_s \tan \theta)]}$$

8. Pseudo Force & Non-Inertial Frames (JEE Focus)

Inertial Frame: Frame moving with constant velocity ($a = 0$). Newton's laws apply directly.

Non-Inertial Frame: Accelerating frame ($a \neq 0$). Newton's laws apply only after adding a **Pseudo Force**:

$$F_{pseudo} = - m a_{frame}$$

Directed opposite to the frame's acceleration, acting at the center of mass of body m .

9. Sample Solved Problems (Board & JEE Level)

Problem 1 (CBSE Board Level - Pulley System)

Two masses $m_1 = 3 \text{ kg}$ and $m_2 = 5 \text{ kg}$ are connected by a light inextensible string passing over a smooth pulley. Find the acceleration of the system and string tension. (Take $g = 10 \text{ m/s}^2$)

Solution:

Using Atwood formulas:

$$a = [(m_2 - m_1) / (m_1 + m_2)] g = [(5 - 3) / (5 + 3)] \times 10 = (2 / 8) \times 10 = 2.5 \text{ m/s}^2$$

$$T = [2 m_1 m_2 / (m_1 + m_2)] g = [2 \times 3 \times 5 / 8] \times 10 = 37.5 \text{ N}$$

Problem 2 (JEE Mains Level - Inclined Plane with Friction)

A block of mass 2 kg rests on a rough plane inclined at 30° . If coefficient of static friction $\mu_s = 0.6$, determine whether the block will slide, and calculate the force of friction acting on it.

Solution:

1. Component of weight down incline: $F_{\text{down}} = mg \sin 30^\circ = 2 \times 10 \times 0.5 = 10\text{ N}$.

2. Normal Force: $N = mg \cos 30^\circ = 2 \times 10 \times (\sqrt{3}/2) = 10\sqrt{3} \approx 17.32\text{ N}$.

3. Maximum Static Friction available: $f_{s,\text{max}} = \mu_s N = 0.6 \times 17.32 = 10.39\text{ N}$.

Since $F_{\text{down}} (10\text{ N}) < f_{s,\text{max}} (10.39\text{ N})$, the block does **NOT** slide.

Because static friction is self-adjusting, $f_s = F_{\text{down}} = 10\text{ N}$.